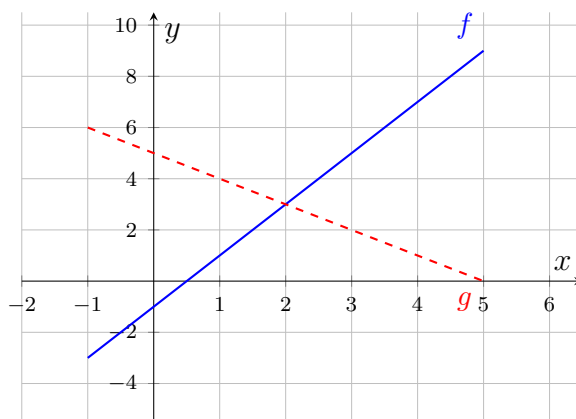


Evaluating Functions from Rules, Tables, and Graphs

Grade 8 / Pre-Algebra

Evaluating a function from its rule, from a table of values, and from a graph, then comparing two function values at the same input.

- $f(3)$ means “the output of f when the input is 3” — it is never f times 3.
- Substitute inside parentheses first, then simplify.
- Problems 5, 7, 8, 11, and 12 all read from **Graph A** below. Every other problem carries its own rule or its own table.



Graph A. The solid line is $y = f(x)$; the dashed line is $y = g(x)$. Read values off the grid.

- 1.** The rule for a function is $f(x) = 3x + 4$.
Find $f(2)$.

Answer: _____

- 2.** The rule for a function is $f(x) = x^2 - 5$.
Find $f(-3)$. Be careful with the sign before you square.

Answer: _____

3. The table gives every value of the function h that was recorded.

x	1	2	3	4	5
$h(x)$	7	10	13	16	19

Find $h(4)$.

Answer: _____

4. The rule for a function is $g(x) = \frac{x+7}{2}$.
Find $g(5)$.

Answer: _____

5. Use **Graph A**. Find $f(3)$ by locating the input on the horizontal axis and reading the height of the solid line above it.

Answer: _____

6. The table gives every value of the function p that was recorded.

x	0	1	2	3	4
$p(x)$	8	5	4	5	8

For which input is the output of p the smallest? Write that input value.

Answer: _____

7. Use **Graph A**. Find $g(1)$ — read the dashed line, not the solid one.

Answer: _____

8. Use **Graph A**, whose rules are $f(x) = 2x - 1$ and $g(x) = -x + 5$.

Evaluate both functions at the input 1, then write $<$ or $>$ so the sentence is true:

$$f(1) \text{ ______ } g(1)$$

Answer: _____

- 9.** The rule for a function is $f(x) = 5 - 2x$.
Find $f(-4)$.

Answer: _____

- 10.** A club charges a joining fee plus a monthly fee. The table shows the total cost C , in dollars, after n months.

n	1	2	3	4	5	6
$C(n)$	12	19	26	33	40	47

How many dollars more is $C(6)$ than $C(2)$?

Answer: _____

Evaluate both functions at the input 4, then write $<$ or $>$ so the sentence is true:

Answer: _____

Explain, in two or three sentences, why both rules give the same output at that input, and what the crossing point tells you about comparing f and g on either side of it. Support your explanation with values from the graph or the rules.